



Machine learning algorithms: paradoxical adjuvants for determining bail and making decisions?

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Introduction

One thing that challenges judges is how they made decisions before trials when artificial intelligence was not yet ubiquitous. These pretrial decisions help people get out of jail while waiting for their trial, and they have to believe the idea that everyone is innocent until proven guilty. But it is hard to know if someone will break the law or commit another crime while they are waiting for trial. The whole idea is that rules and judicial standards are too vague, leaving trial judges with a lot or less power to make sapient bail determinations and decisions. This can lead to unfair and inaccurate bail determinations and decisions that may undermine judicial process. Basically, trial judges decide how risky someone is based on things like missed hearings or trials, or if they are going to reoffend while they are waiting for trial. However, these assessments are not based on any real numbers or statistical data, which can make them less reliable as one literature suggests (Morin-Martel 2023). Trial judges in court often rely on a general rule or their own hunch gavel-to-gavel.

In the context of the digital transformation of the Philippine judiciary, there is a need to create a framework that uses machine learning algorithms to assist judges in fair and accurate bail determinations and decisions.

Evolution of Philippine bail system

Let us first explore the history of our bail system. The Philippines has a long history, dating back to the Spanish colonial era. In those days, a criminal trial started with a complaint filed by the prosecution. Once the complaint was made, a process called a “sumario” began. During this time, the lawyers involved in the case conducted their own investigations and built up evidence for both sides. In essence, the sumario was like a preliminary inquiry that happened before the defendant was officially charged. Sometimes, the sumario could take years, as the defendant was still in prison. If the penalty for

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the crime was prison correccional or more than six months, there was no bail system in place. After the sumario, the trial would finally begin, and the defendant would remain in prison.

When the Americans took control, Major-General Otis created General Orders No. 58, also known as the Criminal Procedure for the Philippines. He then gathered skilled lawyers and justices who were familiar with Spanish laws to draft a new set of rules. The Code came into effect on May 15, 1900. One important part of the Code was the introduction of the writ of habeas corpus, a right guaranteed in the U.S. Constitution. This allowed people to challenge their detention or imprisonment and be released if it was illegal. (Rossabi 1997)

The 1935 Constitution laid the groundwork for bail.² People can be released on bail before a conviction is announced, as long as they have enough surety. But this does not apply if they are charged with a capital offense and there is strong evidence against them. Basically, it makes sure bail is not too high, so people are not stuck in jail waiting for their trial. This rule was also put in place in 1973.³ But the 1987 Constitution,⁴ for the first time, provided provisional release on recognizance, subject to an enabling law.⁵

The bail process in the Philippines has evolved gradually since Otis' General Orders No. 58. After the end of American colonization on July 4, 1946, the criminal procedure underwent several revisions: the 1964 Rules of Court, the 1985 Rules on Criminal Procedure, the 1988 Rules on Criminal Procedure, and the 2000 Revised Rules of Criminal Procedure, as amended. Under these adaptable procedures, any grant or denial of bail depends on the potential penalties.

Debate over bail algorithm

To address the judicial quagmire, many jurisdictional courts now deploy algorithmic risk assessment tools to aid judges in weighing these risks. They use computer programs or machine learning algorithms to help them determine and decide who gets bail. Its use has spread to 60 jurisdictional courts across the United States (Fine and Marsh 2024). These tools look at a

² Const. (1987), art.III, §16 (Phil.).

³ Const. (1973), art. IV, §18 (Phil.).

⁴ Const. (1987) art. III, §13 (Phil.).

⁵ Recognizance Act of 2012, Rep. Act. No. 10389 (Mar. 14, 2013) (Phil), elibrary.judiciary.gov.ph.



lot of things about the defendant, like their criminal history, job history, mental health history, where they live, and other important details. Then, the trial judge determines and decides how risky the defendant is and whether they should be released on bail.⁶ Our jurisdictional courts, however, do not use machine learning algorithms for risk assessment to date or expect to anytime soon.

Untangling artificial intelligence, machine learning, and algorithms can be spiny. Technological jargons are even more complex than legal terms. Artificial intelligence is a set of technologies that gives computers the capability to solve problems and do things that usually need human brains or help.⁷ While machine learning is a set of different ways to teach computers to learn on their own, so they can get better and better at what they do over time.⁸ Machine learning is like mimicking the human brain. An algorithm, on the other hand, is a special mathematical formula that tells software what to do with data.⁹ The good thing about algorithms is that they are really good at predicting things in real time where accuracy depends on the quality and volume of data. They could help predict if someone will flee or get into trouble again. These predictions could help judges decide how much bail to set.

Is machine learning algorithm a double-barreled assessment tool?

There are four relevant streams of literature on machine learning algorithms used in bail decisions. These studies all share a common theme — they support the relational theory of procedural justice. This theory emphasizes fairness in the process rather than the legal result (Tyler 1990). Here are a few interesting findings:

1. The judicial process itself has inherent value, not just the outcomes. (Waldron 2011).
2. People value fairness in the process more than getting the right legal result. These factors include respect, trustworthiness, neutrality, and voice (Meyerson et al. 2020).

⁶ Brook Hopkins and Colin Doyle, *The Pathways of Pretrial Reform*, 57 ABA JUDGES' J. 31 (2018).

⁷ James E. Baker, Laurie N. Hobart. Matthew Mittelsteadt, *An Introduction to Artificial Intelligence for Federal Judges*, www.fjc.gov.

⁸ Id.

⁹ James E. Baker, Laurie N. Hobart. Matthew Mittelsteadt, *An Introduction to Artificial Intelligence for Federal Judges*, www.fjc.gov.



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3. Even if a defendant gets the right legal result, if the procedure was not fair, it could hurt their self-esteem and make them less likely to follow the law (Sorial 2020).
4. Machine learning algorithms can work well with “trustworthiness” and “voice” as criteria; judges can make their decisions more trustworthy by using machine learning as a tool, as long as they have control over the final bail decisions; if there is a conflict between how accurate the algorithm is and how transparent it is, the judge, the human reasoner, could give more weight to “algorithmic accuracy” over transparency; and future use of a Judicial Assistance System or JAS as a support tool on bail decisions can achieve a “more actual and rich trustworthiness towards defendants” (Morin-Martel 2023).

One thing that comes out of these literatures is the fairness in the process, especially the “trustworthiness” component (sincerity of decision makers) and “voice” component (individual’s opportunity of expression of views) (Meyerson et al. 2020).

In the context of Philippine bail practices, impacted individuals indeed are more concerned about the reason behind the application’s denial rather than the outcome itself. Filipinos possess an innate sensitivity (De Leon, Jr. 2011). They would always air or share their concerns with the court or someone else. The fairness of the process is therefore their paramount concern, not the outcome. Trust is the fundamental principle upon which fairness is built. Without it, impacted individuals may perceive a sense of deprivation and a violation of their rights.

Despite the marvels of algorithms, some legal scholars are worried that the factors considered in some algorithm models are limited, assumption-based, less accurate, and heighten biases such as diversity issues, raising concerns about their trustworthiness (Morin-Martel 2023).

Proponents of machine learning algorithms used in bail assessments face two major constitutional hurdles: due process and equal protection. Machine learning algorithms’ “black boxes” raise distinct due process and equal protection concerns when parties are not permitted to examine an algorithm’s parameters and weights. Another literature cautions that when designing artificial intelligence systems, it is crucial to think about the specific



situation in which they will be used, experts in the relevant field should be consulted, and black boxes should not be used to make decisions that involve liberty or legal rights, like in criminal cases (Meltzer 2022). Machine learning companies might not share the secret code behind their algorithms, so impacted individuals cannot test and question them. Question after question lingers.¹⁰

- May a defendant meaningfully challenge the logic of an algorithm if the source code is kept from them? Is it enough for them to have access only to the inputs and outputs the algorithm processes and generates but not the decisional framework it uses?
- May a defendant meaningfully challenge an algorithmic risk assessment without access to its training, testing, and real-world-use data?
- If the algorithm uses machine-learning, and no one, not even the developer, understands its “analysis,” can courts ensure due process of law?
- How many courts test algorithms for accuracy, especially when they predict future (i.e., unrealized) human behavior?

Algorithms, which contain and produce social and other biases, raise another equal protection concerns. MTI Technology Review is sounding an alarm bell — “[u]sing historical data to train risk assessment tools could mean that machines are copying the mistakes of the past”.¹¹

Additionally, “overfitting” is another potential equal protection issue. Overfitting is a phenomenon where a machine learning model becomes overly proficient in the data it has been trained on, failing to adequately address variations or uncertainties. Consequently, biased results may emerge. The machine learning model should possess sufficient flexibility to accurately interpret data that it has not previously encountered, in so doing mitigating the occurrence of this phenomenon. If not, it will raise equal protection concerns (Baker et al. 2022).

¹⁰ James E. Baker, Laurie N. Hobart. Matthew Mittelsteadt, *An Introduction to Artificial Intelligence for Federal Judges*, www.fjc.gov.

¹¹ Karen Hao, *AI Is Sending People to Jail—and Getting It Wrong*, MIT Tech. Rev. (Jan. 21, 2019), <https://www.technologyreview.com/s/612775/algorithms-criminal-justice-ai/>.



Although all these concerns are subject to debate, they may still cause a sense of unease among individuals, particularly those who are technophobic. Despite these technicalities and constitutional hurdles, there are promising machine learning algorithms that may come along and fine-tune these concerns towards increased reliability.

Judges as gatekeepers

Judges play a crucial role in machine learning algorithms' adaptation. First, they serve as the guardians of the law, safeguarding fundamental principles such as the Bill of Rights, established regulations, and evidentiary standards. They become indispensable gatekeepers of algorithmic risk assessment tools. Second, as potential algorithm users, they decide whether to trust algorithm results for bail, among other things. They have to believe that machine learning algorithms can only “predict, not conclude” (Baker et. al. 2023). One literature suggests that public trust in artificial intelligence can be influenced by the judges' trust in the artificial intelligence use. Judges' embrace of artificial intelligence can influence public opinion and acceptance of artificial intelligence (Fine and Marsh 2024). In other words, the public tends to place greater trust in bail algorithms when judges utilize them.

Closing thoughts

In conclusion, three compelling arguments were presented. First, bail machine learning algorithms have both positive and negative consequences, making them a paradoxical algorithmic assessment tool. While we acknowledge the progress made in technology, it is crucial to recognize that certain fundamental rights may need to be compromised to fully harness the benefits of technology. The shibboleth win some, lose some might just apply to this situation. Despite these concerns, there are compelling reasons to anticipate the emergence of promising machine learning algorithms that may address these concerns and ultimately strike a harmonious balance between fundamental rights and technological breakthroughs.

Second, to obtain fair and accurate bail assessments, it is crucial to integrate trust value into consideration, given the unique and sensitive perspective of Filipinos towards processes. If we use a bail algorithm like the one shown here, it should help prevent unfair and inaccurate bail



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determinations and decisions. This is something that courts and algorithm providers should profoundly consider.

And third, it is crucial to clarify that machine learning algorithms are not intended to supplant human judges. Algorithms should only be used as adjuvants to bail determinations and decisions. In the event of conflict, judges' determinations and decisions should be the final authority. As illustrated in this context, judges assume dual roles within the realm of artificial intelligence, primarily serving as guardians of the law and fairness in the process.

As courts gradually embrace artificial intelligence, Supreme Court Chief Justice Alexander G. Gesmundo cautioned on over-applications of artificial intelligence, emphasizing that "courts should still be populated with human judges if we are to expect that empathy and compassion would still go into the consideration of granting equitable remedies."¹² Indeed, judges should be human beings, and we should never replace them.

¹² Chief Justice Alexander G. Gesmundo, AI Can Assist Judges. But Should Never Replace Human Elements of Justice (Oct. 13, 2024), <http://sc.judiciary.gov.ph>.



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